

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

Roll No.:T001	Name:Soham Acharekar
Class:TYIT	Batch:1
Date of Assignment:4/8/26	Date/Time of Submission:4/8/26

2. a. Develop a PWA that works in offline mode using Service Worker.

CODE:

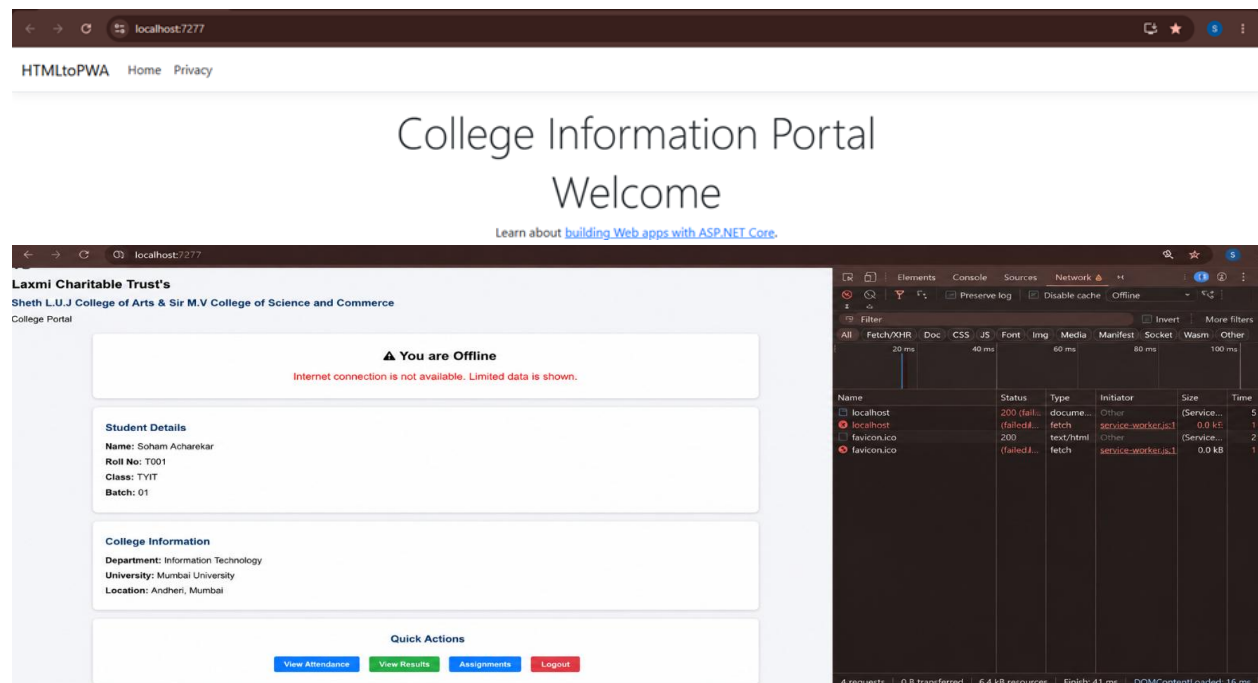
offline.html :

```
<!DOCTYPE html>
<html>
<head>
<title>College Portal</title>
<style>
body {
font-family: Arial; background: #f4f6f9; margin: 0;
}
.header {
background: #003366; color: white;
text-align: center; padding: 10px;
}
.container { width: 80%; margin: auto;
margin-top: 20px;
}
.card {
background: white; padding: 15px; margin-top: 15px; border-radius: 8px;
}
h3 {
color: #003366;
}
button {
padding: 8px 15px; margin: 5px; border: none; color: white; cursor: pointer;
}
.blue {
background: #007bff;
}
.green {
background: #28a745;
```

```
}
.red {
background: #dc3545;
}
.offline { color: red;
text-align: center;
}
</style>
</head>
<body>
<div class="header">
<h2>Laxmi Charitable Trust's</h2>
<h3>Sheth L.U.J College</h3>
<p>College Portal</p>
</div>
<div class="container">
<div class="card">
<h2 style="text-align:center;">⚠ You are Offline</h2>
<p class="offline">No internet connection</p>
</div>
<div class="card">
<h3>Student Details</h3>
<p>Name: Soham Acharekar </p>
<p>Roll No: T001</p>
<p>Class: TYIT</p>
<p>Batch: 01</p>
</div>
<div class="card">
<h3>College Info</h3>
<p>Department: IT</p>
<p>University: Mumbai University</p>
<p>Location: Andheri</p>
</div>
<div class="card" style="text-align:center;">
<h3>Actions</h3>
<button class="blue">Attendance</button>
<button class="green">Results</button>
<button class="blue">Assignments</button>
<button class="red">Logout</button>
</div>
</div>
</body>
</html>
```

service-worker.js :

```
const CACHE = "v1"; self.addEventListener("install", event => {
  event.waitUntil(
    caches.open("offline-cache").then(cache => { return cache.addAll([
      "/",
      "/offline.html"
    ]);
  });
  console.log("Service workers Installed ");
});
self.addEventListener("fetch", event => { event.respondWith(
  fetch(event.request).catch(() => { return caches.match("/offline.html");
  })
);
});
```

OUTPUT:**b. Create a Student Record Management PWA.****CODE:****HomeController :**

```
using Microsoft.AspNetCore.Mvc;
using Practical6.Models; using System.Diagnostics;
using WebApplication8.Models; namespace Practical6.Controllers
{
  public class HomeController : Controller
  {
    public IActionResult Index()
```

```

{
student s = new student(); s.RollNo = 101;
s.Name = "Vinayak"; return View(s);
}
public IActionResult Privacy()
{
return View();
}
[ResponseCache(Duration = 0, Location = ResponseCacheLocation.None, NoStore = true)] public
IActionResult Error()
{
return View(new ErrorViewModel { RequestId = Activity.Current?.Id ?? HttpContext.TraceIdentifier });
}
}}

```

index.cshtml :

```

@{
ViewData["Title"] = "Home Page";
}
<div class="text-center">
<h1 class="display-4">College Portal</h1>
<p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web apps with ASP.NET
Core</a>.</p>
<h1>Offline Page</h1>
<h2>Student Record</h2>
<p>Roll No : @Model.RollNo</p>
<p>Name : @Model.Name</p>
</div>

```

Student.cs :

```

namespace WebApplication8.Models
{
public class student
{
public int RollNo { get; set; } public string Name { get; set; }
}
}

```

service-worker.js :

```

const CACHE = "v1"; self.addEventListener("install", e => {
e.waitUntil( caches.open(CACHE).then(cache => {
return cache.addAll([ "/",
"/Home/Index", "/css/site.css", "/js/site.js"
]);
})
);
});

```

```
self.addEventListener("fetch", e => { e.respondWith(  
  caches.match(e.request).then(response => { return response || fetch(e.request);  
  })  
});
```

OUTPUT: